

Solving Mixed Strategy Equilibrium Using Machine Learning: Evidence from Stock Exchange Fees

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Abstract

Stock exchanges compete for order flow by setting “make” fees for limit orders and “take” fees for market orders. No pure-strategy Nash equilibrium exists in this two-dimensional fee competition, as adjusting the (make, take) fee structure attracts different segments of the market. We provide the first solution for the mixed-strategy equilibrium using neural networks. Our results show that competition improves overall market functioning by increasing transaction volume. However, one side of the market captures most of the gains. Our method can also be applied to other two-dimensional competitive settings, such as games in which both price and quality vary continuously.

1 Introduction

Nash equilibrium provides a standard framework for analyzing how multiple decision makers interact strategically and identify strategy profiles from which no player can gain by unilaterally deviating. Although much of the literature has focused on single-dimensional price competition, many real-world markets involve competition in two-dimensional decisions.

Stock exchanges provide a clear example of such competition. Equity trading occurs on electronic limit-order book venues, where an exchange’s customer base consists of liquidity makers and takers. A liquidity maker initiates a transaction by submitting a limit order—an offer to buy or sell stock at a specified price. Execution occurs when a liquidity taker accepts the posted offer using a market order. Exchanges compete for order flow by setting

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two-sided transaction fees: a “make” fee for liquidity-providing limit orders and a “take” fee for liquidity-removing market orders. The exchange that executes the trade earns revenue from the total fee, calculated as a sum of the make and take fees. A worse cumulative take-side price (i.e., including take fees) makes the offer less attractive to takers. As a result, when a maker chooses an exchange for posting an order, she must consider both the make fee (analogous to “price”) and the likelihood of execution (analogous to “quality”), which depends on the take fee.

This represents two-dimensional competition in the purest sense. There are no frictions or search costs. In fact, the SEC goes as far as to describe the exchanges as “cloned” markets SEC (2015a). Yet, the SEC (2015a) describes that “competitive pricing often leads [stock] exchanges to modify their pricing frequently, typically on a calendar-month basis”. Why do stock exchanges adjust fees so frequently and how does that impact market functioning?

Chao et al. (2019) show that two-dimensional price competition in make-take fee structures precludes the existence of any pure strategy equilibrium. First, no pure strategy equilibrium exists if any exchange earns a positive profit. The intuition is similar to standard Bertrand competition, except that the price competition now operates along two dimensions. Consider the exchange earning the highest positive profit; a competitor can match its take fee and offer a slightly better make fee to capture the entire order flow. Second - and more surprisingly - a pure strategy equilibrium does not exist even when all exchanges earn zero profit. In this case, an exchange can still deviate by adjusting the composition of fees to appeal to one side of the market and earn positive profit. After ruling out the possibility of pure strategy equilibrium, Chao et al. (2019) prove the existence of a mixed strategy equilibrium but are unable to explicitly characterize it. Our paper fills this gap by characterizing the mixed strategy equilibrium using machine learning methods.

The core difficulty arises from the structure of the equilibrium as a fixed point in an infinite-dimensional space. Each firm chooses a probability distribution over a continuum of make-take fee combinations, and a mixed-strategy equilibrium requires that each firm’s distribution be a best response to the other’s. In other words, every action in the support of the distribution must yield the same profit, and that profit must be higher than for any action outside the support. The challenge is that the support is unknown, and only a small subset of distributions can be described analytically. There is no reason to expect the equilibrium to fall within this smaller set. As a result, traditional analytical or computational methods offer limited traction in characterizing the equilibrium.

In this paper, we develop a new machine learning method to solve the mixed strategy problem by combining two key insights. First, mixed strategy equilibria are hard to solve

but easy to verify once a candidate solution is obtained. Mathematically, if distribution f is a symmetric mixed strategy equilibrium, neither exchange can improve its expected payoff by unilaterally deviating from f . We can easily compute the profits from deviating using numerical integration and Monte Carlo simulation. We assume Exchange 2 chooses distribution f . Given that assumption, we can calculate Exchange 1's profits from responding optimally and expected profits from also choosing f . If f is indeed a symmetric equilibrium, the two values must be equal.

Second, we approximate the space of all probability distributions f , i.e. functions mapping make-take fee pairs to probabilities, using neural networks. We ensure the function is positive and integrates to 1 through the neural network architecture and normalization. This transforms the fixed-point problem into finite-dimensional optimization over the neural network's parameters. Given that neural networks are universal function approximators Hornik et al. (1989), they provide a natural nonparametric approach for modeling the distribution over strategies. This is particularly useful in our setting, because existence guarantees allow use to formulate a loss function with a known minimum value. This feature is uncommon in most machine learning applications, where the target function is often unknown or observed with noise.

Using our novel algorithm, we solve for the mixed strategy equilibrium and explain how competition changes the marketplace compared to a monopoly baseline. Firm profits are squeezed but still positive. Furthermore, in contrast with monopoly, duopolistic firms pricing strategies have a strong temporal factor: makers are heavily compensated because they act first. If makers do not pick an exchange, the exchange is guaranteed zero profit. In addition, competition increases market functioning. Both maker and taker expected surplus increases, while exchange profits are significantly lowered. In markets with competition, the probability of a successful transaction is much higher than in a monopoly, where the operator runs the same number of exchanges ¹.

1.1 Related Literature and Contributions

Our paper is the first to characterize the equilibrium exchange fee structure under competition with discrete pricing. When traders can quote continuous prices, Colliard and Foucault (2012) show that the fee structure is irrelevant – a result analogous to the tax neutrality principle in economics: given a fixed total tax, it does not matter whether the burden falls on producers (liquidity makers) or consumers (liquidity takers). If an exchange increases the fee charged to makers while reducing the fee to takers – holding the total fee constant – a liquidity maker

¹A monopolist runs multiple exchanges to engage in price discrimination Chao et al. (2019)

can simply adjust her quoted price to offset the fee change. As a result, exchanges compete only on total fees, and Bertrand competition drives profits to zero as liquidity makers flock to the exchange offering the lowest total fee.

Subsequent research focused on discrete tick pricing, which better reflects regulatory constraints in actual stock markets. Discreteness breaks fee neutrality: traders can no longer fully offset fee changes if they are smaller than one tick. The existing literature on discrete pricing has characterized only monopoly settings (e.g., Foucault et al. (2013); Chao et al. (2019); Riccò et al. (2024)), due to the technical challenges of analyzing two-dimensional price competition. Chao et al. (2019) establish only the existence of a symmetric mixed-strategy equilibrium. In contrast, our paper is the first to fully characterize the equilibrium pricing strategy in this two-dimensional setting.

More generally, we inform the literature on price dispersion. Here, there is an important distinction between 1D and 2D price competition. Traditional works on 1D competition require frictions to prevent consumers from flocking to the market with lowest price Rosenthal (1980), Varian (1980), Baye et al. (2006). Specific frictions are studied further including network externalities Rochet and Tirole (2011) and Li et al. (2020); search costs Spulber (1996); and switching costs Cantillon and Yin (2011).

However, 2D price dispersion can occur without any frictions because prices can not be uniformly ranked. The two dimensional nature occurs naturally in two-sided markets and platform economics, but require additional restrictions to find a solution. For example, Fabra and Montero (2022) only provide an explicit mixed strategy characterization for a special case of their model. Many platform economics models focus on network externalities and pricing strategies to prevent entry, see Jullien et al. (2021) and Xue et al. (2020). Instead, we show how intrinsically homogeneous exchanges compete for order flow in (price, quality) space without any frictions.

To solve for the mixed strategy equilibria, we introduce a general methodology using machine learning. While general existence of Nash Equilibria is guaranteed by results Chao et al. (2019) and Dasgupta and Maskin (1986), they give little insights of the explicit characterization. Mixed strategies in general are hard to analyze, and analytical distributions are unknown except for a few toy examples such as those in Fudenberg and Tirole (1991). As explicitly stated in Tadelis (2013), “For games with many players, or with two players who have many strategies, finding the set of mixed-strategy Nash equilibria is a tedious task”. Our paper introduces a natural way to use machine learning to solve for mixed strategy equilibria.

It applies to problems beyond standard algorithms for finding mixed strategy equilibria,

which are often restricted to discrete action spaces and have exponential runtime complexity, see Nisan et al. (2007) for an overview. They cannot be applied to our problem because we have a continuous action space and any fine approximation via discretization will have long runtimes. Furthermore, our model lacks common features that make optimization problems easier, i.e. monotonicity, convexity, and continuity making quadratic programming solutions infeasible.

We also extend the relatively new literature of using machine learning to solve analytically intractable problems in finance and economics. We are the first to do so in a mixed strategy setting. Fernández-Villaverde et al. (2021), Fernández-Villaverde et al. (2023), and Azinovic et al. (2022) use neural networks to solve for dynamic equilibrium models, but do not include the game theory aspects of mixed strategy equilibria. Wu and Jaimungal (2023) and Duarte et al. (2024), use neural networks to model the actions of multi-period agents, i.e. given a state, what is the best response for an agent. We use neural networks purely as a function approximator so that we can apply gradient descent.

Our work also provides a new insight for approximating functions in the finance literature. Previous works such as Winschel and Krätzig (2010) and Winberry (2018) use a basis of polynomials. Neural networks have an advantage because they handle nonconvexities extremely well LeCun et al. (2015), a necessary factor in solving our model. Furthermore, the high-dimensional optimization problem can be solved efficiently through a combination of gradient descent and back-propagation. This setup allows us to use hundreds of parameters to get a better approximation for the mixed strategy distribution.

2 Institutional Details

All stock exchanges in the US are for profit organizations, and earn money through exchange fees. Regulation National Market System (Reg NMS) governs the market microstructure of all exchanges and includes rules promoting the national market system. Under the electronic limit order book structure, a trader can post a limit order to an exchange and become a liquidity maker or submit a market order and become a liquidity taker. Liquidity makers agree to fill an order at their specified price but with unknown execution probability. Liquidity takers act against posted limit orders, and orders are filled instantly. Such services are not free, however. Exchanges charge a make and take fee per share for each side of the transaction.

Rule 610, the access rule, prevents exchanges from giving preferential treatment to members/subscribers over non-members/non-subscribers. This reduces switching costs, rules out network effects, and greatly reduces cost of entry.

Rule 612, the Sub-Penny Rule, sets the minimum price increment of NMS stocks to be 1 tick. For stocks priced above \$1, a tick is \$0.01. This divides a stock’s effective price into two parts. Market participants quote at discrete tick levels, while exchanges adjust the continuous, sub-penny component through make-take fees. Makers and takers have no control over sub-penny price shifts as they are entirely determined by exchange fee structure.

Despite the sub-penny nominal value of the fees, large transaction quantities make the fees non-trivial. In fact, these fees serve as a major source of revenue, accounting for 34.7% of NASDAQ’s revenue in 2012 O’Donoghue (2015), and the big three stock exchanges even sued the SEC over trading fee caps Osipovich and Michaels (2019).

3 Model

We extend the model proposed in Chao et al. (2019). A trade is modeled as a three-period game among the following risk neutral parties: two stock exchanges, a liquidity maker, and a liquidity taker.

We assume prices are normalized on the domain $[0, 1]$, which represents the price between one tick. Without loss of generality, we assume liquidity makers are buyers and liquidity takers are sellers. Let the maker’s valuation v_b follow a uniform distribution on $[\frac{1}{2}, 1]$, and the taker’s valuation v_s follow a uniform on $[0, \frac{1}{2}]$ ². This guarantees that all parties want to make a trade, but are limited by the discrete nominal price. A trade occurs when cum fee prices are accepted by both the maker and taker.

The buyer and seller must trade through an exchange and post prices at 0 or 1 (the ticks). Actual (cum fee) prices are determined by quoted price plus any fees or rebates. In the model, two fee structures can lead to same cum-fee price. Suppose the exchange wants to target a cum fee make price of p and a cum fee take price t . The exchange can give rebates of $1 - p$ to the maker and charge $1 - t$ fee to the taker. Makers will quote a price of 1. If the trade is successful the maker buys the stock at $1 - (1 - p) = p$. Takers sell the stock at $1 - (1 - t) = t$. No makers will quote a price of 0 because they know the taker will not agree to sell the asset for $0 - (1 - t) = t - 1 < 0$.

Similarly, the exchange can charge a maker fee of p and give a taker rebate of t to achieve the same cum fee price. The maker submits an order at 0, and buys the stock for a cum fee price of $0 + p = p$. The taker sells the stock for $0 + t = t$. In this case, no makers will quote a price of 1 because they will pay $1 + p > 1$ to buy the stock.

Due to the equivalence of fee structures, we focus only on cum-fee prices p and t . Figure

²This assumption simplifies analysis, but is not necessary for our algorithm.

1 shows the actions available to each party in each period. At Time 0, all exchanges simultaneously decide on the make/take fees to maximize expected profits. We model the cum fee (buy, sell) price at Exchange i , denoted by (p_i, t_i) ³

At Time 1, nature draws on liquidity maker at random with valuation v_b . After liquidity maker observes prices, the liquidity maker submits a limit order to Exchange i to maximize expected surplus

$$BS^i := (v_b - p_i)P(v_s \leq t_i)$$

If $BS^i < 0$ for all i , the liquidity maker does not submit any order. $(v_b - p_i)$ represents the gains from execution, and $P(v_s \leq t_i)$ represents the probability that the taker will accept the offer.

At Time 2, nature draws a liquidity taker with valuation v_s . The taker cannot pick exchanges because only one exchange has a limit order. They can, however, choose to reject trading if their surplus is negative. Exchanges make profits only when the trade is successful, with profits given by the difference between make and take prices $p_i - t_i$.

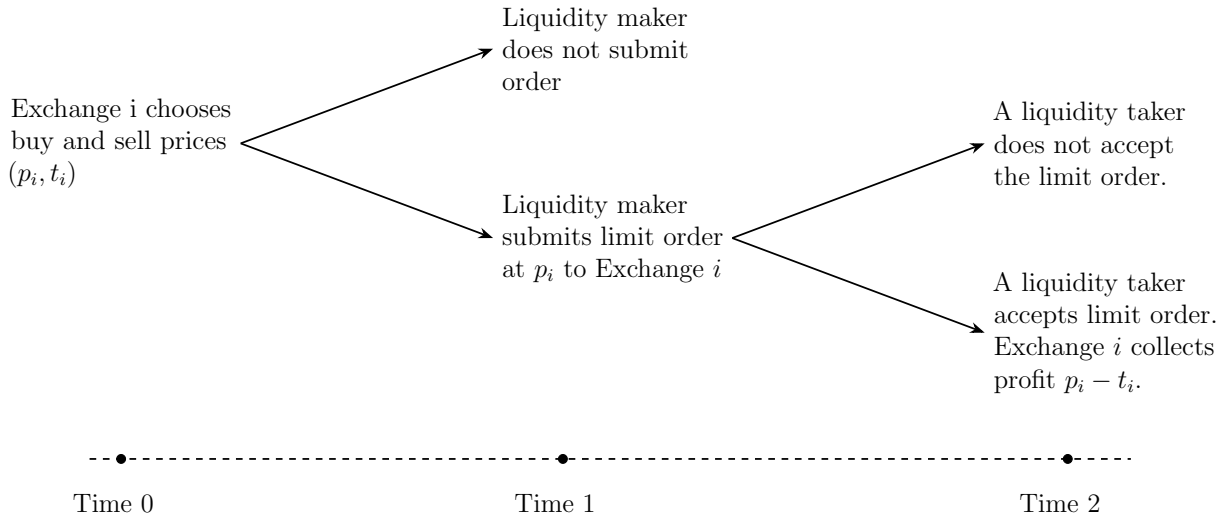


Figure 1: Timeline of events.

The taker's decision also offers an alternative interpretation of fees within the model. From the maker's perspective, the take price t_i serves as a proxy for exchange quality. A higher t_i implies a lower cost to the taker, which increases the likelihood that the maker's limit order will be executed. Conversely, submitting an order to an exchange with a low t_i raises the

³Due to Rule 612 (discrete tick sizes), participants cannot neutralize fees, and exchanges have full control over price within a tick, see Chao et al. (2019).

chance that the order remains unfilled.

3.1 Analysis

We solve for the equilibrium characterization of the sequential-move game through backward induction. Let $\Theta(v_b)$ and $\Phi(v_s)$ represent the cdf of the maker's and taker's valuation.

Liquidity Taker: The liquidity taker's choice is simple, conditional on t_i , they accept the trade if and only if $v_s \leq t_i$. The probability that the taker accepts is $\Phi(t_i)$.

Liquidity Maker: Conditional on p_i, t_i, p_j, t_j , the liquidity maker submits to Exchange i if

$$(v_b - p_i)\Phi(t_i) \geq (v_b - p_j)\Phi(t_j) \quad (1)$$

$$v_b \geq p_i \quad (2)$$

The first equation compares the expected utility of choosing Exchange i with choosing Exchange j , while the second ensures that submitting an order does not lead to negative payoffs. The maker only enters the market when $v_b > \min(p_i, p_j)$.

The maker's decision rule is shown in Proposition 1.

Proposition 1. *Let*

$$\gamma = \frac{p_i\Phi(t_i) - p_j\Phi(t_j)}{\Phi(t_i) - \Phi(t_j)}$$

when it is well defined i.e. $\Phi(t_i) \neq \Phi(t_j)$. The probability that a random maker picks Exchange i given that prices are p_i, t_i, p_j, t_j

$$P(\text{Maker picks Exchange } i \mid p_i, t_i, p_j, t_j) = \begin{cases} 1 - \Theta(\max(\gamma, p_i)) & \text{if } t_i > t_j \\ \max\{\Theta(\gamma) - \Theta(p_i), 0\} & \text{if } t_i < t_j \\ [1 - \Theta(p_i)] \mathbb{I}(p_i < p_j) & \text{if } \Phi(t_i) = \Phi(t_j)^4 \end{cases}$$

where $\mathbb{I}(\cdot)$ represents the indicator function.

All else being equal, investors prefer higher quality t_i and lower price p_i , but they face a price-quality trade off when both (p_i, t_i) change. In general, high valuation makers choose high price and high quality while low valuation makers choose low price and low quality. γ is the cutoff. Figure 2 shows the probability of choosing Exchange i for various values of p_i, t_i when (p_j, t_j) is fixed.

⁴This is not the same as $t_i = t_j$. In particular, if both exchanges offer prices so that the taker always accepts, it is possible for $t_i \neq t_j$, but $\Phi(t_i) = \Phi(t_j)$.

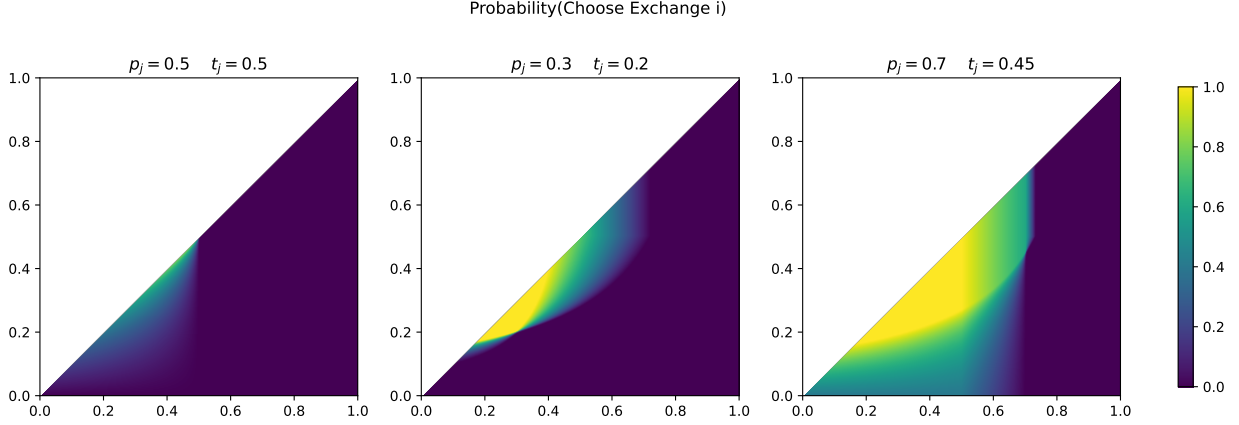


Figure 2: Plots of the probability that a maker chooses Exchange i conditional on Exchange j 's choice. Brighter areas correspond to price-quality pairs that result in Exchange i having a higher chance of being picked by the maker.

The complex structure of Proposition 1 begins to shed light on why a pure strategy Nash Equilibrium does not exist. For example, when $(p_j, t_j) = (0.3, 0.2)$, there are incentives for Exchange i to decrease p_i by a tiny ϵ while keeping quality the same. Doing so, decreases profits by ϵ , but guarantees that Exchange i will be picked. Even if exchanges offered $p_j = 0.5, t_j = 0.5$ – which will process all trades under a single exchange monopoly – makers with extreme high/low valuations will prefer higher/lower prices and quality.

Exchange i : Given the probability of participation from the above sub-games and conditional on Exchange j choosing (p_j, t_j) , Exchange i 's expected profit is

$$\pi(p_i, t_i \mid p_j, t_j) := (p_i - t_i) \cdot \Phi(t_i) \cdot P(\text{Maker picks Exchange } i \mid p_i, t_i, p_j, t_j)$$

Figure 3 shows $\pi(p_i, t_i \mid p_j, t_j)$ for different (p_i, t_i) when (p_j, t_j) is fixed. Notice that high probability of being selected does not always translate to higher profits. This is because a higher likelihood of selection means a lower p_i and a higher t_i squeezing profits.

4 Mixed Strategy Equilibrium

There is no pure-strategy equilibrium for this game Chao et al. (2019). Similar to Bertrand competition, pure-strategy equilibrium cannot exist with positive profits. Suppose Exchange 2 is making more positive profits than Exchange 1. Then Exchange 1 can undercut Exchange 2's make price by a tiny $\epsilon > 0$ and mimic the take price. This steals all of Exchange 2's business which outweighs the loss incurred from lower net fees.

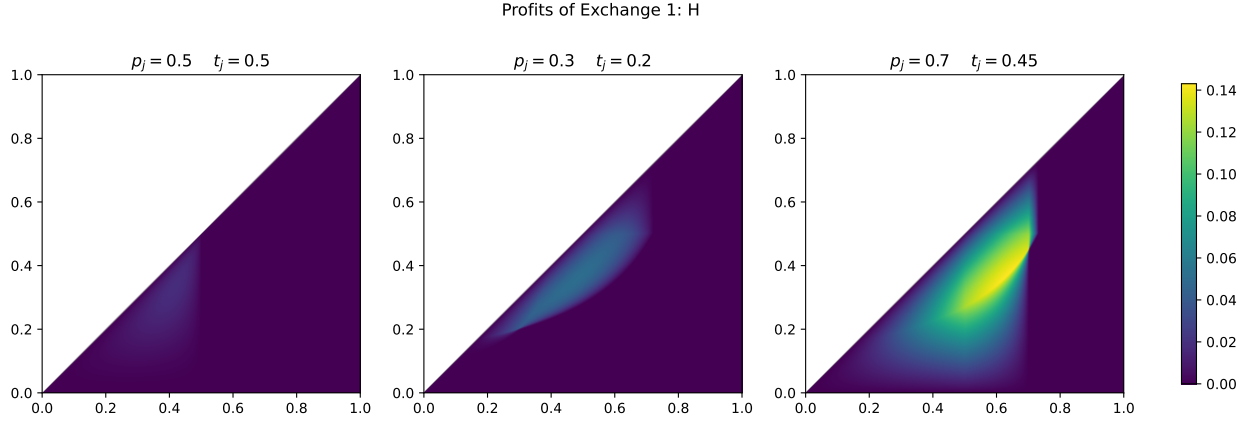


Figure 3: Plots Exchange i 's profit conditional on Exchange j 's choice. Brighter areas correspond to price-quality pairs that result in more profits for Exchange i .

Even with both exchanges making zero profits, a pure strategy equilibrium still does not exist. This follows from Proposition 1. For each $p_j = t_j \in [0, 1]$, there always exists a (p_i, t_i) with $p_i > t_i$ such that

$$P(\text{Maker picks Exchange } i \mid p_i, t_i, p_j, t_j) > 0$$

Again, the intuition is that some investors prefer high price, high quality exchanges whereas others prefer low price, low quality exchanges.

Chao et al. (2019) prove the existence of a symmetric mixed strategy equilibrium, but does not provide a characterization. Exchanges choose a fee structure from a probability distribution function $f(p, t)$ with the support $\Omega := \{(p, t) \mid f(p, t) > 0\}$.

Lemma 1. *The support Ω is a subset of the triangle in $\mathbb{R}^2$ given by $\Lambda := \{(p, t) \mid p \leq 1, t \geq 0, p \geq t\}$*

Note that Exchange i will never pick $p_i < t_i$ because it has negative payoff regardless Exchange j 's choices. Furthermore, $p_i > 1$ and $t_i < 0$ is never a best response because the maker/taker never chooses Exchange i , and profits are guaranteed to be 0.

Finding the exact distribution analytically is difficult. First, $\pi(p_i, t_i \mid p_j, t_j)$ is non-convex and non-smooth. This means that there are no convergence guarantees for even standard maximization problems on $\pi(p_i, t_i \mid p_j, t_j)$, let alone an integral of $\pi(p_i, t_i \mid p_j, t_j)$ arising from expectations. More importantly, we do not know the functional form of $f(p, t)$. The space of functions is an infinite dimensional, and general optimization problems on function spaces are not well studied. Lastly, standard algorithms involving discretization quickly become computationally intractable. Discretizing each dimension into N points, yields a N^4 grid.

For these reasons, we solve for the mixed strategy equilibrium using neural networks.

4.1 Equilibrium No Deviation Condition

We now first derive a necessary and sufficient condition that is satisfied in equilibrium. Assume that Exchange j chooses the probability distribution $f(p, t)$. Then Exchange i 's expected profit from choosing (p_i, t_i) is

$$\begin{aligned}\Pi(p_i, t_i) &:= E_f [\pi(p_i, t_i \mid p_j, t_j)] \\ &= \int_{\mathbb{R}^2} \pi(p_i, t_i \mid p_j, t_j) f(p_j, t_j) dp_j dt_j\end{aligned}$$

where we use $E_f[\cdot]$ to denote expectations with respect to the probability density $f(p_j, t_j)$.

Conditional of Exchange j 's strategy, Exchange i 's expected profit is a function of its own price-quality pair. To maximize profits, Exchange i needs to solve

$$\max \Pi(p_i, t_i) = \max E_f [\pi(p_i, t_i \mid p_j, t_j)] \quad (3)$$

Chao et al. (2019) proves there exists a symmetric equilibrium, in which case Exchange i also chooses $f(t, p)$. Exchange i 's expected profit from this strategy is

$$E_f[\Pi(p_i, t_i)] = \int_{\mathbb{R}^2} \Pi(p_i, t_i) f(p_i, t_i) dp_i dt_i \quad (4)$$

$$= \int_{\mathbb{R}^4} f(p_i, t_i) \pi(p_i, t_i \mid p_j, t_j) f(p_j, t_j) dp_j dt_j dp_i dt_i \quad (5)$$

In equilibrium, Exchange i must have no incentives to deviate from $f(p, t)$. Mathematically, this means that its expected payoff $E_f[\Pi(p_i, t_i)]$ and its best action payoff $\max \Pi(p_i, t_i)$ are equal. This gives us the *no deviation condition* on $f(p, t)$.

$$\max \Pi(p_i, t_i) - E_f[\Pi(p_i, t_i)] = 0 \quad (6)$$

Equation 6 has a simple economic explanation. Let (p_i^*, t_i^*) be a maximizer to Equation 3. A mixed strategy exists because because the maximizer (p_i^*, t_i^*) is not unique. In fact, for Equation 6 to hold, every fee structure in the support Ω must give the same payoff. This makes exchanges indifferent between realizations in Ω . Furthermore, any fee structure outside of Ω yields smaller or equal profits than (p_i^*, t_i^*) , removing any incentives for exchanges to offer alternative fee structures.

To find $f(p, t)$ we minimize over the space of distributions $\mathcal{D}$,

$$\min_{f \in \mathcal{D}} \left(\max \Pi(p_i, t_i) - E_f[\Pi(p_i, t_i)] \right) \quad (7)$$

5 Numerical Algorithm

We now present an algorithm to implement the minimization problem in 7.

5.1 Neural Networks

This section covers the basics of neural network architecture, and how it relates to our solution. For a general overview of machine learning, see Goodfellow et al. (2016).

To approximate the equilibrium probability distribution, we use a fully-connected feed-forward neural network ⁵. The basic architecture is shown in Figure 4. Neural networks are organized by layers consisting of nodes. The input layer (green) takes in (p, t) and maps it to a value $f(p, t)$ in the output layer (red) through a series of linear transformations and function compositions. Any layers in between (green) are called hidden layers. They keep track of intermediate values in the calculation but are not presented with the final result.

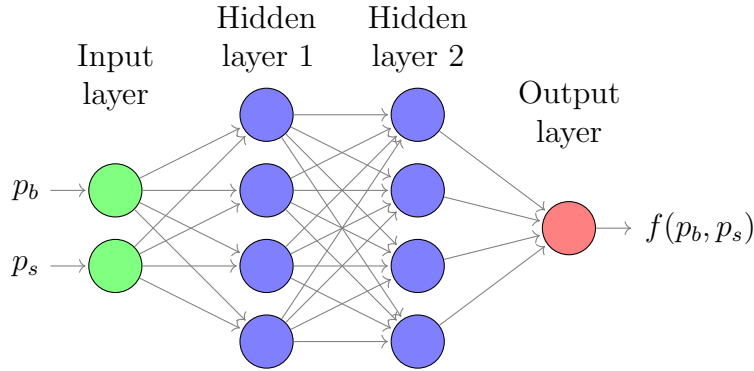


Figure 4: A Fully Connected Feed Forward Neural Network with 2 hidden layers consisting of 4 nodes each. Each circle represents a node, and arrows indicate how information flows through the network. For example, the Output Layer takes in all values from Hidden Layer 2, but does not take in any values from Hidden Layer 1.

Consider any blue or red node in Figure 4. Let $\mathbf{X}_n$ represent the inputs to node n , $\mathbf{W}_n$ be a vector with the same dimension as $\mathbf{X}_n$, and b_n be a constant. The output of the node is

$$\zeta(\mathbf{W}_n^T \mathbf{X}_n + b_n)$$

⁵Fully-connected means that each node is connected to all nodes in the previous layer. Feed forward means that values are passed from one layer to the next in a chronological order.

The weights $\mathbf{W}_n^T$ and b_n are unique to each node, and $\zeta(z)$ is an activation function. For our setup, we use SiLU activation functions, i.e. $\Phi(z) = \frac{z}{1-e^{-z}}$ throughout the neural network.

Training the neural network amounts to finding $\mathbf{W}_n$ and b_n that best approximate $f(p, t)$. This is possible due to the universal approximation theorem, Hornik et al. (1989) and Bach (2017), which states that neural networks can reasonably approximate any unknown Borel measurable function.

Figure 4 only provides a simplified illustration. For actual computations, we use 5 hidden layers each with 10 nodes. The additional layers and nodes yield a better approximation, but, like all machine learning algorithms, our method cannot escape the bias-variance trade-off: more complicated models tend to overfit. The impact of the trade-off is smaller than those in empirical works because the trade-off only exists when sample size is fixed. In our simulation framework, sample size can be adjusted as needed and the only real constraint is runtime. We examine this in more detail in Section 8 by running convergence simulations based on sample size.

5.2 The Algorithm

We use a neural network to solve for $f(p, t)$. This machine learning method solves the continuous domain problem because the neural network maps any $(p, t) \in \mathbb{R}^2$ to $\mathbb{R}$. To evaluate the quantity on the left side of Equation 6, we use Monte Carlo simulation similar to Longstaff and Schwartz (2001) and solve the minimization problem 7 using gradient descent.

In each iteration, we first generate a random vector of actions

$$\mathcal{A}^T := [a_1, a_2 \dots a_N]$$

by uniformly sampling from the triangle defined in Lemma 1. Each a_k contains a fee structure (i.e. $a_k \in \mathbb{R}^2$). While $\mathcal{A}$ represents a two dimensional space, we write it as a vector. It is the most efficient way to store the discretization because matrices imply a rectangular support, which we know is not true.

We simulate a new $\mathcal{A}$ at every iteration in our algorithm. This makes our setup different from discretization on a grid, and prevents the neural network from learning the particularities related to a discretization.

We assume that both exchanges can only select actions within $\mathcal{A}$. Given the action set $\mathcal{A}$, we can now find the discrete analogue to $\pi(p_i, t_i, p_j, t_j)$. Let $\boldsymbol{\pi}$ be a $N \times N$ matrix representing the profits for all N^2 possible action combinations. In particular, let the (k, m) entry of $\boldsymbol{\pi}$ represent the profit Exchange 1 gets from choosing action a_k and Exchange 2 choosing

a_m , respectively. In other words, each column $\boldsymbol{\pi}_k$ contains all possible profit realizations if Exchange 1 chooses action a_k . This is a vector because Exchange 1's profits depend also on Exchange 2's actions.

Both exchanges' strategies are represented by the same neural network that takes in each action a_k and outputs a number f_k to describe the probability of an exchange choosing that action. This gives a vector of probabilities

$$\mathbf{f}^T := [f_1, f_2, \dots, f_N]$$

One neural network suffices in this case, because we are considering symmetric equilibria. For now, we assume that $\mathbf{f}^T$ is a probability distribution, and explicitly enforce this constraint later. Again, $\mathbf{f}^T$ represents a function in two dimensional space, but we write it as a vector. From a notation standpoint, this makes the representation consistent with matrix multiplication which we use below.

To get the analogue of $\Pi(p_i, t_i)$, the profits in $\boldsymbol{\pi}_k$ must be weighted by the likelihood of Exchange 2 picking each actions. Thus, expected profits for Exchange 1 from choosing action a_k are given by the vector

$$\begin{bmatrix} f_1 & f_2 & \dots & f_N \end{bmatrix} \begin{bmatrix} \pi_{11} & \pi_{12} & \dots & \pi_{1N} \\ \pi_{21} & \pi_{22} & \dots & \pi_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{N1} & \pi_{N2} & \dots & \pi_{NN} \end{bmatrix} = \begin{bmatrix} \mathbf{f}^T \boldsymbol{\pi}_1 \\ \mathbf{f}^T \boldsymbol{\pi}_2 \\ \vdots \\ \mathbf{f}^T \boldsymbol{\pi}_N \end{bmatrix} \quad (8)$$

Each entry in the vector represents Exchange i 's expected profit from picking i while Exchange j randomizes its fee according to $\mathbf{f}$. The optimal action is simple: the firm picks an action that gets $\max_k \mathbf{f}^T \boldsymbol{\pi}_k$ in profits.

In a symmetric equilibrium, Exchange 1 also picks mixed strategy $\mathbf{f}$, and earns expected profit $\mathbf{f}^T \boldsymbol{\pi} \mathbf{f}$. The exchange has no incentives to deviate if $\mathbf{f}^T \boldsymbol{\pi} \mathbf{f}$ is equal to the best action's profit given by $\max_k \mathbf{f}^T \boldsymbol{\pi}_k$.

Alternatively stated, $\max_k \mathbf{f}^T \boldsymbol{\pi}_k - \mathbf{f}^T \boldsymbol{\pi} \mathbf{f}$ can be used to approximate the continuous No Deviation Condition in Equation 6 . Define

$$\mathcal{L}(f) := \left(\max_k \mathbf{f}^T \boldsymbol{\pi}_k - \mathbf{f}^T \boldsymbol{\pi} \mathbf{f} \right)^2$$

as the loss function. Finding the minimizer for $\mathcal{L}(f)$ yields a mixed strategy equilibrium.

The choice of squared loss is analogous to linear regression which minimizes the squared residual error. The quadratic function, being differentiable, often leads to nicer convergence in training.

We minimize $\mathcal{L}(f)$ by performing gradient descent on the parameters of f . Recall from Section 5.1 f is completely determined by its parameters. We initialize the neural network with random parameters θ_0 , and determine subsequent parameters θ by

$$\theta \leftarrow \eta \nabla_{\theta} \mathcal{L}(f)$$

where η is the learning rate selected using ADAM, see Kingma and Ba (2014). Intuitively, one can think of gradient descent as rolling a ball down a landscape. The gradient step updates in the direction of steepest descent similar to how a ball would roll down a hill under the influence of gravity. converge to a local minima. Evaluating the gradients of $\mathcal{L}(f)$ requires backpropagation through the neural network, a technique that is well-developed in the machine learning community.

A summary of the steps is shown in Algorithm 1.

Algorithm 1 Solving mixed strategy equilibrium

- 1: **Inputs** Select N (sample size in Monte Carlo simulations) and T (number of iterations in gradient descent)
 - 2: **Start:** Initialize a neural network f with random parameters to represent the probability distribution.
 - 3: **for** $i = 1, 2, \dots, T$ **do**
 - 4: Generate random actions in 2 dimensional space $\mathcal{A}^T = [a_1, a_2, \dots, a_N]$.
 - 5: Calculate payoffs $\boldsymbol{\pi}$ for all action pairs.
 - 6: Calculate probabilities $\mathbf{f}$, where the i -th entry of $\mathbf{f}$ is $f_i = f(a_i)$
 - 7: Compute loss $\mathcal{L}(f) := (\max_k \mathbf{f}^T \boldsymbol{\pi}_k - \mathbf{f}^T \boldsymbol{\pi} \mathbf{f})^2$
 - 8: Update the neural network parameters to minimize $\mathcal{L}(f)$ using gradient descent.
 - 9: **end for**
-

5.3 Constraints on $\mathbf{f}$

As mentioned previously, $\mathbf{f}$ needs to be a probability density function. This means that the neural network must always output positive values and the density must sum to 1.

We build the first constraint into the neural network architecture by applying a softplus activation function on the neural network's output node. The softplus function is defined as $\Phi(z) = \ln(1 + e^z)$ and guarantees that outputs are positive.

The second constraint, $\sum_{k=1}^N f_k = 1$ is difficult to enforce directly. We can add a penalty

term as in Azinovic et al. (2022)

$$\mathcal{L}(f) + \lambda \left(\sum_{k=1}^N f_k - 1 \right)^2 \quad (9)$$

This setup does not change the minimizer. Any distribution that results in a mixed strategy equilibrium will yield the minimum loss of zero. However, the penalty term changes the learning landscape for gradient descent. In particular, $\mathbf{f} = \mathbf{0}$ is a minimizer for $\mathcal{L}(f)$, but not a probability distribution. Gradient steps of $\mathcal{L}$ point towards $\mathbf{0}$, but the penalty term prevents the function from going to $\mathbf{0}$. As a result, little progress is made when implemented.

We adopt an alternative approach by normalizing the outputs of the neural network using

$$f_k / \sum_{k=1}^N f_k$$

This removes $\mathbf{0}$ from the optimal solution set and “levels the landscape”. One tradeoff with normalization is that outputs of our neural network will be proportional to a probability density distribution. All relative probabilities are correct, but to achieve a proper density function, we need scale by numerically integrating the mass under the curve.

6 Two Dimensional Monopoly

Before presenting our results on the mixed strategy distribution, we first summarize results from monopoly in Chao et al. (2019) to better contrast between competition and monopoly. We allow the monopolist to open more than one exchange. This is consistent with real markets as there are more than 10 exchanges but only 3 main holding companies (NASDAQ, Intercontinental Exchange, and BATS Global Markets).

Lemma 2. *Only prices within $\frac{1}{2} \leq p_i \leq 1$ and $0 \leq t_i \leq \frac{1}{2}$ for all exchanges i are rational.*

Lemma 2 arises because the monopolist never overcompensates makers or takers. Full maker participation occurs at 0.5, so offering $p < 0.5$ only harms profits. Similarly, at $t = 0.5$, all takers will accept a posted limit order. Any higher take prices do not benefit the exchange. Instead, the monopolist adopts a menu pricing as shown in Proposition 2.

Proposition 2. *For a monopolist operating K exchanges, the optimal make-take fee pair (p_i, t_i) for Exchange $i \in \{1, \dots, K\}$ is*

$$p_i = \frac{1}{2} + \frac{i}{2(2K+1)} \quad t_i = \frac{i}{2K+1} \quad (10)$$

The expected surplus of the liquidity maker, liquidity taker, and monopoly operator, respectively, is

$$MS(K) = \frac{K(K+1)}{3(2K+1)^2}, \quad TS(K) = \frac{K(K+1)}{3(2K+1)^2}, \quad \Pi(K) = \frac{2K(K+1)}{3(2K+1)^2}$$

Liquidity makers are segmented into $K+1$ groups depending on their quality preference through second order price discrimination: higher valuation makers will choose higher quality exchanges. Notice that Exchange i charges the “mark-up” of $\frac{1}{2} - \frac{i}{2(2K+1)}$ for providing services.

The probability that a maker chooses Exchange i is $2\left(\frac{K+i+1}{2K+1} - \frac{K+i}{2K+1}\right)$, and the probability that a taker accepts the order is $\frac{2i}{2K+1}$. Summing across all exchanges, we get that the probability of successful trade is

$$4 \sum_{i=1}^K \left(\frac{K+i+1}{2K+1} - \frac{K+i}{2K+1} \right) \cdot \frac{i}{2K+1} = \frac{2K(K+1)}{(2K+1)^2}$$

Proposition 2 indicates that the expected surplus for all participants is increasing in the number of exchanges. Exchanges take half the surplus, while makers and takers evenly split the remaining half. This is a surprising result because more price discrimination benefits all the players. The increase in welfare arises through increased maker participation. With K exchanges, $\frac{1}{2K+1}$ of the makers do not submit a trade, but this fraction decreases with more exchanges.

As K approaches to infinity, the lowest make price approaches $\frac{1}{2}$, and almost all liquidity makers will participate. The probability of a successful trade approaches $\frac{1}{2}$. In a monopoly, exchange fees cause a barrier to trading. While every maker and taker would be better off after a trade, even with infinite exchanges only half of the potential trades are executed.

7 Two Dimensional Competition

We use our algorithm to solve for the equilibrium under competition. We run our algorithm with $N = 40000^6$ and perform $T = 500$ gradient iterations. The algorithm converges to a loss of $\mathcal{L}(f) \approx 5.6 \times 10^{-7}$. This loss should be considered with respect to the maximum and minimum values for $\mathcal{L}(f)$. The minimum value is 0, and only the true equilibrium solution can achieve this value. The maximum loss is bounded by monopoly profits. This occurs when both Exchange choose $(p, t) = (0, 0)$ with probability 1. Exchange i 's current profit is 0, but

⁶This can be interpreted as approximately a 200×200 grid although points will be randomly distributed.

can achieve the monopoly profit of $\frac{4}{27}$ if it deviates to the monopoly solution. The loss for the $(0, 0)$ strategy is $\left(\frac{4}{27}\right)^2 \approx 2.19 \times 10^{-2}$. We examine convergence of the algorithm in more detail in Section 8.

Figure 5 shows the statistical distribution that characterizes the mixed strategy equilibrium. A higher resolution can be obtained by increasing the number of samples N and adding more layers to the neural network, but our results already capture the key economic insights.

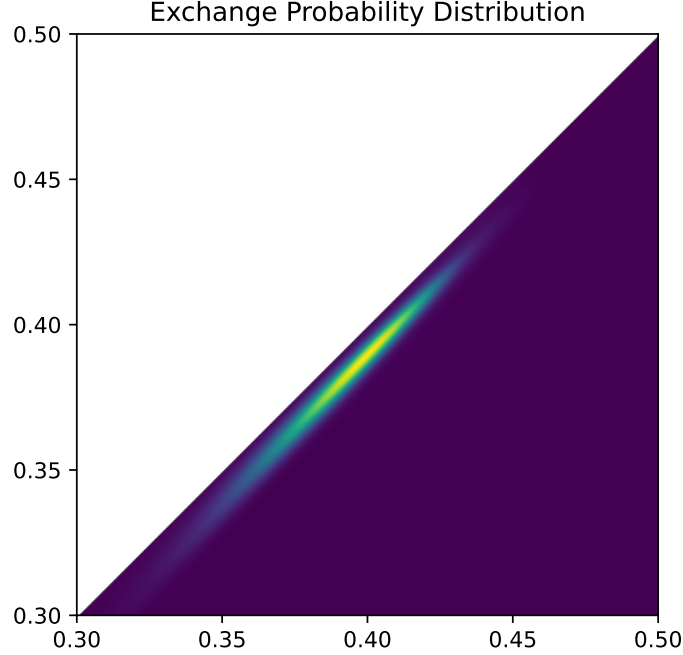


Figure 5: The equilibrium distribution of both exchanges. The x -axis represents p_i and the y -axis represents t_i . Brighter areas correspond the price quality pairs that are more likely to be selected. Areas not shown in the plot but are in the triangle from Lemma 1 have probability zero.

Under competition, exchanges heavily favor the liquidity maker. The range for p_i on the x -axis shows that exchanges offer lower prices than the maker's minimum valuation because exchanges must attract the maker in period 1 to have a chance for profit in period 2. If the maker does not choose the exchange, profits are guaranteed to be 0. This motivates vigorous competition for the maker's order. In contrast, such behavior is irrational in a monopoly because all exchanges are owned by the same holding company. Good prices in one exchange will attract all makers and cannibalize the profits of other exchanges.

For duopoly, the effect of attracting liquidity provision is so strong, that exchange fee structure guarantees makers participation regardless of the realization of v_b – the question is which exchange the maker prefers. Full maker participation can be achieved with only

two competing exchanges. Panel A of Table 1 shows the trading probabilities of competition compared with monopoly. A monopoly operator must open an infinite number of exchanges to achieve full maker participation.

Figure 5 shows that competition benefits low valuation takers. Duopoly exchanges typically offer $t_i > 0.3$, while a monopolist's price menu will contain $t_i < 0.3$ for $K \geq 2$. The expected taker surplus are shown in Table 1. Competition increases expected taker surplus from 0.08 (monopoly with 2 exchanges) to 0.143 (duopoly).

The duopoly surplus for both makers and takers is unattainable through a monopoly operator. In duopoly, makers are heavily compensated and obtain more surplus than the other players combined. Takers are squeezed under duopoly but are still better off. In essence, takers get a smaller slice of a much bigger pie.

No. of Exchanges	Duopoly	Monopoly Operator		
	2	$K = 1$	$K = 2$	$K \rightarrow \infty$
A. Trading Probabilities				
Maker submits order	1.000	0.333	0.600	1.000
Successful trade occurs	0.754	0.444	0.480	0.500
B. Welfare				
Operator	0.007	0.148	0.160	0.1667
Operator (per exchange)	0.0035	0.148	0.080	0
Maker Surplus	0.269	0.074	0.080	0.0833
Taker Surplus	0.143	0.074	0.080	0.0833
Total Surplus	0.420	0.296	0.320	0.333

Table 1: A comparison between duopoly competition and a monopoly operation who establishes K exchanges. Panel A displays the probability that makers submit a limit order and the probability of a successful trade at any exchange. Panel B shows the expected surplus for operators, liquidity makers, and liquidity takers. The methods for calculating welfare in a duopoly are included in Appendix B.1. Values in the table may not sum to total surplus due to rounding error.

Not surprisingly, competition squeezes operator profits. Exchanges choose a make-take price structure such that conditional on a successful trade, it makes near constant profit, i.e. the difference between make-take fees is the same for any exchange in the market. Figure 5

shows that profits are small and only around 0.01. Notice that competition does not lead to zero exchange profits. This is consistent with the results derived in Chao et al. (2019). From Table 1 total expected operator profit for both exchanges in a duopoly is 0.007 with each exchange getting 0.0035 in expectation. This is less than one-tenth of monopoly profits.

Overall, competition increases market performance and liquidity. In a monopoly, at most half of all potential trades are fulfilled. The remaining half is lost due to market inefficiency. In contrast, only two competing exchanges are needed to successfully process 75.4% of incoming trades, and total surplus in the market is 0.420. As a comparison, the maximum amount of surplus that can be extracted from the market is $E[v_b - v_s] = 0.5$. The captured surplus is not simply the expected difference in maker-taker valuation multiplied by the probability of a successful trade, i.e.

$$0.5 \times 75.4\% = 0.377 < 0.420$$

because takers who do not participate have lower gains from trade.

8 Algorithm Convergence

The algorithm’s convergence depends on three components: neural network complexity, Monte Carlo sample size N , and number of gradient descent iterations T . In this section, we empirically examine the trade-offs of each parameter on the final loss.

Compared to traditional neural network problems where optimal loss is unknown, the minimum of $\mathcal{L}(f)$ is 0. The minimum is obtained at the symmetric equilibrium whose existence is guaranteed by Chao et al. (2019) and Dasgupta and Maskin (1986). This means that performance of our neural networks can be evaluated on both relative and absolute scales.

Figure 6 shows the algorithm performance for various sample sizes and iterations. Neural network size is fixed to 5 hidden layers each with 10 neurons. Due to the Monte Carlo nature, each repetition yields slightly different results. For a more accurate representation of performance, for each (sample size, iteration) pair, we repeat Algorithm 1 independently for 10 times and plot the median. When evaluating the loss for plotting, we always use $N = 40,000$ samples.

For fixed neural network complexity, number of iterations is more important than sample size. The algorithm has a sublinear convergence rate which is common in many neural network algorithms, and most of the improvement occurs within the first 300 iterations. For the largest sample size, final loss is around 10^{-6} . Based on the experiments, sample sizes in excess of 5000, do not have a noticeable impact on loss. This means our algorithm can

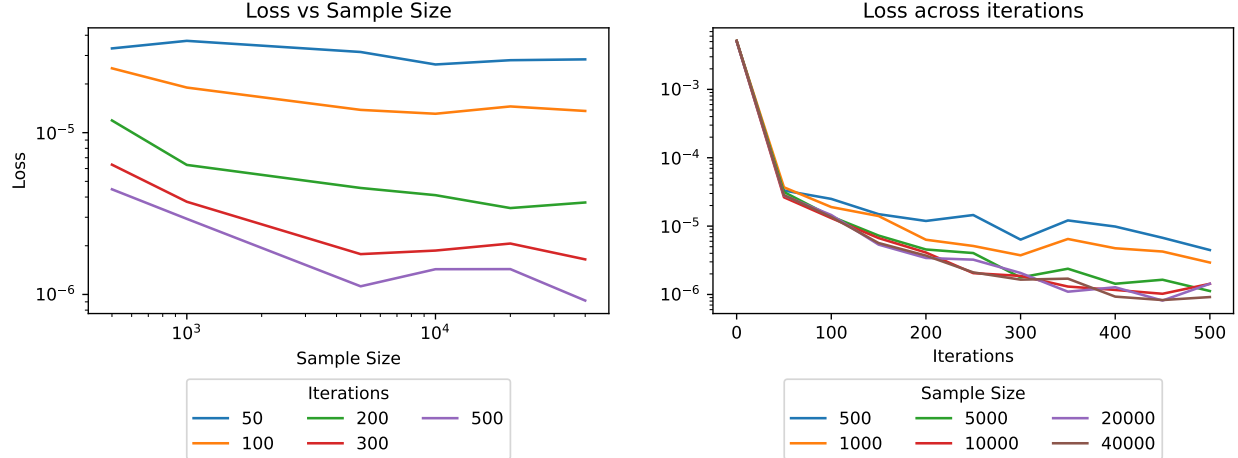


Figure 6: The loss of the algorithm with various sample size and number of iterations. The specifications for these two parameters are shown in the legend. Results are the medians across 10 independent experiments.

be used even when computation resources are constrained. However, sample sizes too small, such as 500^7 , lead to excess variance in the estimator of $\mathcal{L}(f)$ which slows convergence.

In a second set of experiments, we fixed the number of iterations to $T = 500$ and vary neural network complexity. Neural networks are fixed at 10 nodes per layer, but have 1, 2, or 4 hidden layers. Results are shown in Figure 7. Again, we run Algorithm 1 independently 10 times and plot the median. Neural network complexity is an important factor on algorithm performance. Mathematically, neural networks are repeated linear transformations along with function compositions. Additional layers allows the neural network to become closer to a universal function approximator, and this added flexibility is crucial to algorithm performance.

9 Conclusion

This paper characterizes the mixed-strategy equilibrium in a two-dimensional market. Relative to a monopoly setting, competition induces exchanges to favor liquidity makers, reflecting a temporal aspect in transactions—exchanges earn zero profit in the second stage if they are not selected in the first. The result is a shift in surplus toward makers, an increase in the likelihood of successful trades, and an improved overall market efficiency for both makers and takers.

Beyond stock exchanges, the framework applies to a broader class of two-sided markets with similar pricing dynamics. For example, credit card platforms offer rewards to consumers

⁷This is approximately a 22×22 grid.

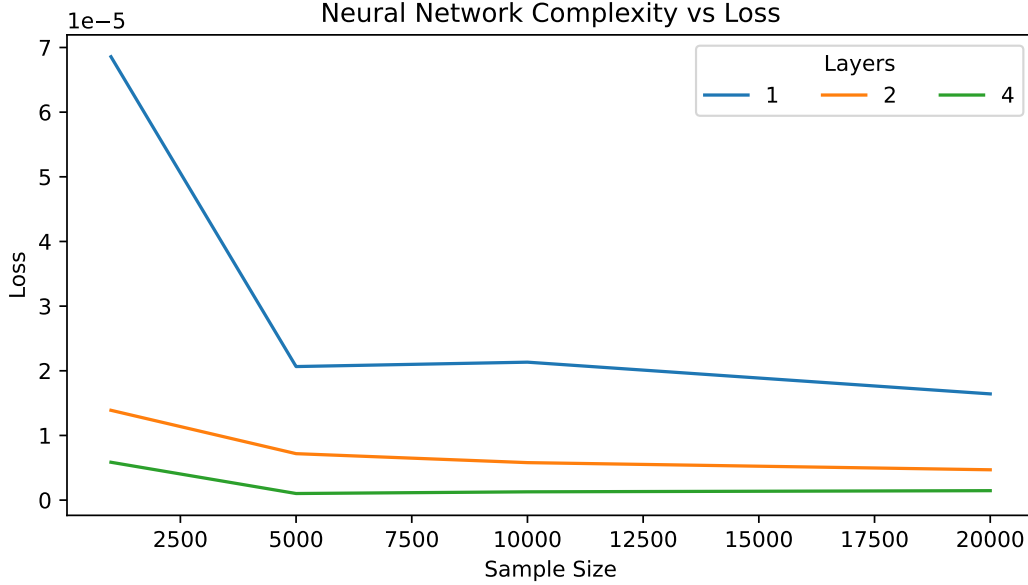


Figure 7: The loss of the algorithm with various sample size and neural network complexity measured by number of hidden layers. Sample size ranges from $\{1000, 5000, 10000, 20000\}$. Results are medians over 10 repetitions.

(analogous to maker rebates) while charging merchants transaction fees. They also frequently change their rewards and fees structures, and there are even cards with rotating cashback categories. This suggest that the credit card equilibrium is also a mixed strategy. Empirical evidence in Bedre-Defolie and Calvano (2013) and Edelman and Wright (2015) show that platforms often over-subsidize the consumer side, a pattern consistent with the incentive structure that emerges in our equilibrium.

From a methodological standpoint, this paper contributes a general approach for solving mixed-strategy equilibria using machine learning. While the equilibrium distribution is analytically intractable, the equilibrium condition is a well-defined —namely, the maximum incentive to deviate must be zero. This makes mixed strategies a natural setting for machine learning, where the functional form of the solution is unknown, but the optimality condition is clear and verifiable. Our method uses neural networks as nonparametric approximators of the probability distribution and reformulates the infinite-dimensional fixed-point problem into a finite-dimensional optimization one. Because existence is guaranteed by economic theory, the learning problem is both interpretable and well-posed. This approach can be extended to a range of settings where optimal price-quality trade-offs lack analytical structure and standard solution methods fall short.

A Appendix: Proposition 1

Rearranging Equation 1 gives that

$$v_b [\Phi(t_i) - \Phi(t_j)] \geq p_i \Phi(t_i) - p_j \Phi(t_j)$$

There are three cases:

1. $\Phi(t_i) > \Phi(t_j)$:

Since the cdf function is strictly increasing, this case occurs when $t_i \in (0, 0.5)$ and $t_i > t_j$. Combined with Equation 2, we get that

$$v_b \geq \gamma \quad v_b \geq p_i$$

The probability of such a v_b is $1 - \Theta(\max(\gamma, p_i))$.

2. $\Phi(t_i) < \Phi(t_j)$:

This occurs when $t_i \in (0, 0.5)$ and $t_i < t_j$. Makers only choose Exchange i when $p_i \leq v_b \leq \gamma$, which occurs with probability $\max(\Theta(\gamma) - \Theta(p_i), 0)$. The maximum ensures that probabilities are not negative in the case that $v_b < p_i$.

3. $\Phi(t_i) = \Phi(t_j)$:

This case occurs when $t_i = t_j$, but also if exchanges offer a price that takers always accept. If $p_i \Phi(t_i) \geq p_j \Phi(t_j)$, then the taker picks Exchange i with probability

$$P(v_b \geq p_j) = 1 - \Theta(p_j)$$

Otherwise if $p_i \Phi(t_i) < p_j \Phi(t_j)$, Exchange i is never selected.

B Appendix: Duopoly Surplus

In this section, we give the numerical procedure to calculate surplus for a duopoly. Assume that both exchanges decide on strategy $f(p, t)$. Let ρ denote the set of price-quality pairs selected by the exchange p_1, t_1, p_2, t_2 , which will partition the support space for our conditional expectations. Let E_i denote the event that the liquidity maker chooses Exchange i .

B.1 Taker Surplus

First fix ρ . Conditioning on ρ guarantees that $0 < t_i < 0.5$ as seen in Figure 5. Using law of total probability,

$$E[\max(t_i - v_s, 0) \mid \rho, E_i] = E[t_i - v_s \mid \rho, E_i, v_s \leq t_i]P(v_s \leq t_i)$$

The distribution of $v_s \mid v_s \leq t_i$ is Uniform on $[0, \frac{1}{t_i}]$, so the above becomes

$$(t_i - \frac{1}{2}t_i) \cdot 2t_i = t_i^2$$

Total taker surplus, conditional on ρ , is

$$\begin{aligned} E[\max(t_i - v_s, 0) \mid \rho] &= E[E(\max(t_i - v_s, 0) \mid \rho, E_i)] \\ &= t_1^2 P(E_1 \mid \rho) + t_2^2 P(E_2 \mid \rho) \end{aligned}$$

All that remains is to integrate $E[\max(t_i - v_s, 0) \mid \rho]$ over ρ . We do this using a Monte Carlo framework. We first simulate an action set $\mathcal{A}$. This is done independently from the neural network training. Since this step does not involve gradient descent iterations, we can use a larger sample size to reduce estimator variance. Define

$$\mathbf{P}_1 = t_1^2 P(E_1 \mid \rho) + t_2^2 P(E_2 \mid \rho)$$

as the matrix evaluated at the points in $\mathcal{A}$. The neural network gives us the probability density $\mathbf{f}$. Thus, maker surplus can be approximated by $\mathbf{f}^t \mathbf{P}_1 \mathbf{f}$.

B.2 Maker Surplus

We proceed similarly by using iterated expectations.

1. $t_i > t_j$:

The condition of choosing Exchange i is equivalent to $v_b \geq \max(\gamma, p_i) = \gamma$. The distribution $v_b \mid v_b \geq \gamma$ is uniform on $[\gamma, 1]$ if $\gamma < 1$. Thus, we have

$$\begin{aligned} E[(v_b - p_i)\Phi(t_i) \mid \rho, E_i] &= \left[\frac{1}{2}(1 + \gamma) - p_i \right] \Phi(t_i) \mathbb{I}(\gamma < 1) \\ &= [(1 + \gamma) - 2p_i] t_i \mathbb{I}(\gamma < 1) \end{aligned}$$

2. $t_i < t_j$:

The event E_i is equivalent to $p_i < v_b < \gamma$. Conditioning on $p_i < v_b$ does not give additional information on v_b since $p_i < v_b$ was guaranteed by the ρ . The distribution of $v_b \mid v_b < \gamma$ is uniform on $[\frac{1}{2}, \min(\gamma, 1)]$.

$$\begin{aligned} E[(v_b - p_i)\Phi(t_i) \mid \rho, E_i] &= \left[\frac{1}{2} \left(\frac{1}{2} + \min(\gamma, 1) \right) - p_i \right] \Phi(t_i) \\ &= \left(\frac{1}{2} + \min(\gamma, 1) - 2p_i \right) t_i \end{aligned}$$

Since f is a continuous probability distribution, $t_i = t_j$ occurs with probability 0, so we do not include the calculations above. Total maker surplus, conditional on ρ , is

$$E[(v_b - p_i)\Phi(t_i) \mid \rho] = \sum_{i=1}^2 E[(v_b - p_i)\Phi(t_i) \mid \rho, E_i] P(E_i \mid \rho) \quad (11)$$

Similar to before, given a trained neural network f , we evaluate Equation 11 on an independently simulated set of actions $\mathcal{A}$ and store it in a 2 dimensional matrix $\mathbf{P}_2$. Taker surplus can be approximated by $\mathbf{f}^t \mathbf{P}_2 \mathbf{f}$.

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